Introduction to Multimodal Learning: Problem Set 3

Motion-Compensated Frame Prediction and Temporal Consistency

1 Motion-Compensated Frame Prediction (Section 1)

We use the synthetic clean-translation clip from `hw3_utils.py`: a 128×128 grayscale sequence in which a 28×28 bright square moves with constant velocity $(dy, dx) = (2, 4)$ pixels per frame. All five consecutive frame pairs $(t-1, t)$ for $t = 1, \ldots, 5$ are evaluated.

1.1 Previous-frame baseline. `frame_difference` returns $\hat{I}_t = I_{t-1}$ and $R_t = I_t - I_{t-1}$. Because the square translates uniformly, the absolute residual concentrates on the leading and trailing edges of the square (Fig. 1), giving MSE = 781.25, PSNR = 19.20 dB and residual entropy 0.158 bits/pixel.

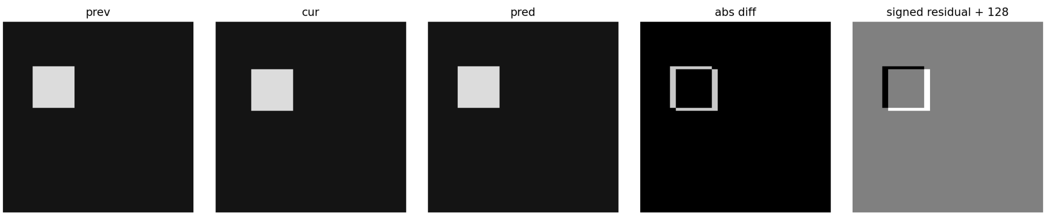

Figure 1: Previous-frame baseline on (t=0, t=1). From left to right: I_{t-1}, I_t, prediction, $|I_t - I_{t-1}|$, signed residual +128. The double-edge structure of the absolute residual is the textbook signature of un-compensated translation.

1.2 Exhaustive block matching. `block_matching` divides I_t into non-overlapping $B \times B$ blocks and exhaustively searches displacements $(u, v) \in [-R, R]^2$ in I_{t-1}, choosing the candidate that minimizes SAD or MSE. The motion field has shape $(H/B, W/B, 2)$ with vectors stored as (dy, dx) such that the candidate is taken from $I_{t-1}[y + dy, x + dx]$. Candidates whose top-left falls outside the frame are rejected. The implementation iterates over the $(2R + 1)^2$ displacements, gathers all $H_b \times W_b$ candidate blocks at each shift, computes per-block costs in one tensor pass, and updates the running arg min via masked element-wise comparisons.

Table 1 averages the four required configurations and the SAD/MSE choice over the five frame pairs of the clean clip. SAD and MSE produce identical motion vectors and identical predictions on this clip (see CSV `result/block_matching_metrics.csv`); we report the SAD column.

Configuration	MSE	PSNR (dB)	H(R_t) (bits/px)	Runtime (s)
Baseline (frame difference)	781.25	19.20	0.158	3.0×10^{-5}
Block matching B=8, R=4	0.00	∞	0.000	0.019
Block matching B=16, R=4	0.00	∞	0.000	0.008
Block matching B=16, R=8	0.00	∞	0.000	0.029
Block matching B=32, R=8	365.62	22.78	0.075	0.017

Table 1: Mean over five frame pairs (SAD metric). For $B \leq 16$ the search radius covers the true displacement $(2, 4)$, giving exact reconstruction. With $B = 32$ a single block straddles the moving square and the static background, so no rigid translation can match it perfectly — a useful illustration of the block-size/motion-coherence trade-off.

The B=16, R=8 run on (t=0, t=1) produces motion vectors that are exactly $(-2, -4)$ over every block touching the moving square and $(0, 0)$ elsewhere, which is the convention of sampling `prev[y+dy,x+dx]` for a square that moved by $(+2, +4)$ in frame t. The vector field and prediction grids are saved as `result/block_b16_r8_mv.png` and `result/block_b16_r8_pred.png` (Fig. 2).

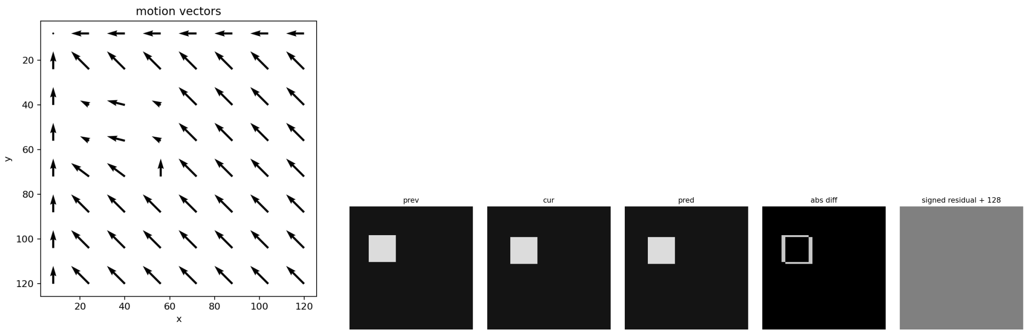

Figure 2: Block matching with B=16, R=8, SAD. Left: motion-vector field, with non-trivial vectors only on blocks containing the moving square. Right: predicted frame and residual; the residual is identically zero.

Failure case. For B=32, R=8 four of the sixteen 32×32 blocks straddle the moving-square / background boundary. Because that boundary is not rigidly translated by any single displacement (only the square is), block matching picks $(0, 0)$ for those blocks and the residual concentrates on the leading/trailing edges. PSNR rises to 22.78 dB versus the baseline's 19.20 dB but cannot reach perfect reconstruction — the classical block-size / motion-coherence trade-off.

2 Temporal Consistency Evaluation (Section 2)

2.1 Objective metrics. We implement variance-of-Laplacian sharpness, mean L1 first-order temporal difference, mean L1 second-order temporal acceleration ($\|I_{t+1} - 2I_t + I_{t-1}\|_1$), and mean block-motion magnitude (from Section 1, B=16, R=8).

Clip	Sharpness	Temp. diff	Temp. accel	Motion mag (px)
clean_translation	566.4	3.91	7.81	9.88
flicker	538.8	46.11	91.40	9.43
jitter	566.4	6.43	12.97	10.05
blur	5.1	2.56	2.42	9.07

Table 2: Per-clip objective metrics. Sharpness is variance of Laplacian; temporal-difference and -acceleration are mean L1 over consecutive and three-frame triplets; motion magnitude is mean L2 of block-matching vectors.

2.2 Human evaluation (1–5 ratings).

- **clean_translation:** visual 5, temporal 5, naturalness 5 — clean uniform translation, no visible artifacts.
- **flicker:** visual 3, temporal 2, naturalness 3 — alternating brightness creates strong flicker between odd/even frames.
- **jitter:** visual 3, temporal 2, naturalness 2 — non-uniform vertical and horizontal offsets give a visibly shaky look.
- **blur:** visual 3, temporal 5, naturalness 4 — Gaussian blur reduces sharpness but motion remains smooth.

2.3 Discussion.

1. *Best-aligned metric.* Temporal acceleration is the most informative for our human ratings. The flicker clip's score (91.4) is an order of magnitude larger than the clean clip (7.8) and the jitter clip (13.0), exactly matching the perceived ordering by temporal consistency. The first-order temporal difference is also large for flicker but not by as wide a margin, since uniform translation already produces a non-trivial first difference.
2. *Misleading metric.* Frame sharpness misranks the blur clip: variance of Laplacian collapses to 5.1, suggesting catastrophic quality, but a human rates the blur clip's *temporal* consistency as 5/5 because the motion is smooth — only the spatial frequencies are gone. Sharpness conflates spatial fidelity with overall video quality. Likewise, temporal-difference misleads on the clean clip vs. jitter: their first-difference values (3.9 vs. 6.4) are close even though humans rank them very differently for temporal consistency.

3. *Why one scalar is insufficient.* Generated-video quality is multidimensional: spatial fidelity (sharpness, distortion), temporal consistency (flicker, jitter), motion plausibility (smoothness, semantics), and content correctness (subject/text alignment) are largely independent. A single scalar must average over these axes and so masks failure modes — VBench's design philosophy is exactly this: a panel of axis-specific metrics with separate scores. Two videos with the same averaged score can be perceptually very different (one blurry but smooth, one sharp but flickering), and a single number cannot tell us which.

3 Short Conceptual Questions (Section 3)

3.1 Optical-flow constraint

Brightness constancy: $I(x+u, y+v, t+1) = I(x, y, t)$. First-order Taylor expansion of the LHS around (x, y, t):

$$I(x, y, t) + I_x u + I_y v + I_t \cdot 1 + \mathcal{O}(\|(u, v, 1)\|^2) = I(x, y, t),$$

which to first order gives the brightness-constancy / optical-flow equation

$$\boxed{I_x\, u + I_y\, v + I_t = 0.}$$

Under-constrainedness (aperture problem). This is one scalar equation in two unknowns (u, v) at each pixel. Geometrically the solution lies on a 1D line in flow-space whose normal is $\nabla I = (I_x, I_y)$: only the flow component along the local image gradient (the *normal flow*) is observable, while the component tangent to an isophote — moving along an edge of constant intensity — is invisible to the constraint. To recover both components one must add a regularizer (Horn–Schunck), assume local flow constancy and solve a 2×2 system over a window (Lucas–Kanade), or impose higher-order assumptions.

3.2 Space-time attention complexity

Let $N_s = HW$ tokens per frame, T frames, hidden dimension d. We count attention-matrix and projection FLOPs in big-Θ.

(a) Joint space-time self-attention. All $TN_s = THW$ tokens attend jointly. The attention map is $(THW) \times (THW)$ and the $\mathrm{QK}^\top$/softmax/attention$\cdot$V cost is

$$\Theta\big((THW)^2 d\big) = \Theta(T^2 H^2 W^2 d).$$

QKV projections add $\Theta(THWd^2)$, dominated by the attention term for large THW.

(b) Factorized (spatial, then temporal) attention. *Spatial*: for each of the T frames, attention over HW tokens costs $\Theta((HW)^2 d)$, summed over frames giving $\Theta(TH^2W^2d)$. *Temporal*: at each of the HW spatial positions, attention over T tokens costs $\Theta(T^2 d)$, summed giving $\Theta(HWT^2 d)$. Total

$$\Theta\big(TH^2W^2 d + HWT^2 d\big) = \Theta\big(THW(HW + T)\, d\big).$$

(c) Numerical comparison for $T{=}16, H{=}W{=}32$. $THW = 16{\cdot}1024 = 16{,}384$, so joint attention pays $(16384)^2 \approx 2.68 \times 10^8$ pairwise scores per head. Factorized: spatial $T(HW)^2 = 16 \cdot 1024^2 \approx 1.68 \times 10^7$; temporal $HW \cdot T^2 = 1024 \cdot 256 \approx 2.62{\times}10^5$. The total $\approx 1.7{\times}10^7$ is roughly **16**$\times$ cheaper than joint, because $HW \gg T$ here so $HW + T \approx HW$ and the saving is exactly the factor T. More generally, factorizing avoids the cross-term $T \cdot HW$ inside the attention map and reduces $T^2(HW)^2$ to $T(HW)^2 + (HW)T^2$. Memory for the attention matrix shrinks correspondingly from $\Theta((THW)^2)$ to $\Theta((HW)^2 + T^2)$.

4 Bonus: Hierarchical Block Matching (Option B)

We build a 3-level Gaussian pyramid (5-tap kernel $[1, 4, 6, 4, 1]/16$ applied separably with reflect padding, then $2\times$ subsample), run exhaustive block matching at the coarsest level with $R{=}4$, and refine each finer level by upsampling the motion field (vectors $\times 2$, grid $\times 2$) and searching ± 1 pixel around the prediction. The effective full-resolution search radius is $4{\cdot}4 + 1{\cdot}2 + 1 = 19$ px, comparable to the exhaustive $R{=}8$ baseline.

Method	Mean PSNR (dB)	Mean MSE	Mean runtime (s)
Exhaustive B=16, R=8	∞ (perfect)	0.0	0.029
Hierarchical B=16, 3 levels	35.13	27.19	0.005

Table 3: Hierarchical vs. exhaustive on five frame pairs of the clean translation clip. Hierarchical search is roughly **6**$\times$ faster on 128$\times$128 frames, at the cost of a finite (still high) PSNR. The residual error originates at the coarsest level: after 2$\times$ downsampling the per-frame displacement is only 1 px, which the integer search rounds; subsequent $\pm$1 px refinement cannot recover sub-block translations on the half-resolution grid where the square boundary is mis-aligned.

The full per-frame numbers, the hierarchical motion-vector visualization (`result/hierarchical_mv.png`) and the prediction grid (`result/hierarchical_pred.png`) are saved alongside the exhaustive outputs in `result/`.

Reproducibility. Run `HW3.ipynb` (CPU only, $<$ 1 min). Outputs are saved to `result/`: GIFs of the four clips, prediction/motion-vector plots, and CSVs `block_matching_metrics.csv`, `consistency_metrics.csv`, `human_eval.csv`, `hierarchical_vs_exhaustive.csv`. A sanity check in cell 6 confirms block-matching vectors are exactly $(-2, -4)$ on the moving square, matching the synthetic ground truth.